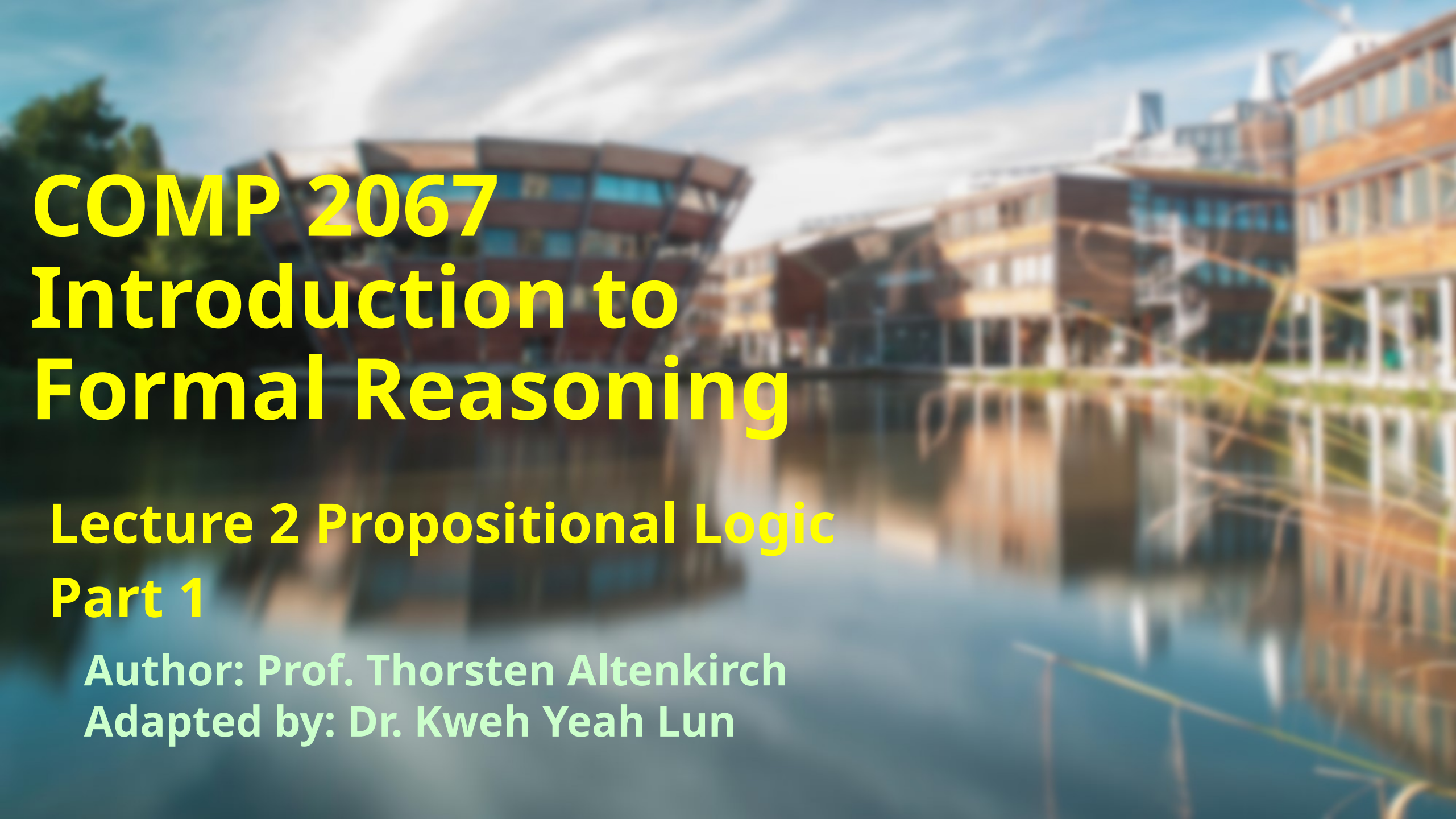




COMP 2067

Introduction to

Formal Reasoning

Lecture 2 Propositional Logic

Part 1

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2.1 Proposition

- **definitive statement** which may be able to prove
- In Lean, **$P : \text{Prop}$** express that P is a proposition
- Declaration in Lean:
 $\text{variables } P \ Q \ R : \text{Prop}$
 - Try it!
 - This means that **$P \ Q \ R$ are propositional variables** which may be substituted by any concrete propositions.
- Examples: “The day is raining” or “We go to UNM”



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Connectives

- Implication ($\rightarrow$): $P \rightarrow Q$
 - If P then Q
- Conjunction ($\wedge$): $P \wedge Q$
 - P and Q
- Disjunction ($\vee$): $P \vee Q$
 - P or Q



- false
 - Example: Kweh can fly.
- true
 - Example: You are UNM student.
- Negation ($\neg$): $\neg P$
 - Not P
- Equivalence ($\leftrightarrow$): $P \leftrightarrow Q$
 - P is equivalent to Q
- $P \leftrightarrow Q$ is defined as $(P \rightarrow Q) \wedge (Q \rightarrow P)$.



- As in algebra, parentheses are used to group logical expressions.
- To save parentheses there are a number of conventions.
- conjunction and disjunction bind stronger than implication and equivalence.
 - $P \vee Q \rightarrow R$ as $(P \vee Q) \rightarrow R$.
- Implication binds stronger than equivalence.
 - $P \rightarrow Q \leftrightarrow R$ as $(P \rightarrow Q) \leftrightarrow R$.
- Conjunction binds stronger than disjunction.
 - $P \wedge Q \vee R$ as $(P \wedge Q) \vee R$.



- Negation binds stronger than all the other connectives.
 - $\neg P \wedge Q$ as $(\neg P) \wedge Q$.
- Implication is right associative.
 - $P \rightarrow Q \rightarrow R$ as $P \rightarrow (Q \rightarrow R)$.
- This is not a complete specification.
- If in doubt use parentheses.



- Proving propositions in Lean
 - If we are proving a statement containing propositional variables then this means that **the statement is true for all replacements of the variables with actual propositions.**
 - It is a **tautology**.
- Tautologies
 - useless in everyday conversations because they contain no information.
 - However, for study of logic, they exhibit the basic figures of reasoning.



2.2 First Proof

- In Lean
 - $p : P$ for p proves the proposition P
- A **proof** is a **sequence of tactics** affecting the current proof state which is the sequence of assumptions we have made and the current goal.
- A proof begins with *begin* and ends with *end* and every tactic is terminated with *,.*



- Let's start with a very simple tautology $P \rightarrow P$: If P then P .
- Example: **If the sun shine then the sun shines.**
- Clearly, this sentence contains no information about the weather, it is vacuously true, indeed it is a tautology.

```
theorem I : P → P :=  
begin  
  assume h,  
  exact h,  
end
```

[Try it!](#)



- *assume h*

- means that prove an implication by **assuming the premise (the left hand side)** and using this assumption to **prove the conclusion (the right hand side)**.
- When you move the cursor before *assume h*, you will see that the proof

```
P : Prop  
⊢ P → P
```

- This means assume that *P* is a proposition and want to prove $P \rightarrow P$.



- The $\vdash$ symbol (pronounced turnstile) separates the assumptions and the goal.
- After *assume* h , the proof state
- This means the goal is now P but there is an additional assumption $h : P$

```
P : Prop,  
h : P  
⊢ P
```



- exact h,
 - **complete the proof** by telling Lean that there is an **assumption that exactly matches the current goal**.
 - If you move the cursor after the , you see *no goals* and it is done.



2.3 Using assumptions

- Next we are going to prove another tautology:

$$(P \rightarrow Q) \rightarrow (Q \rightarrow R) \rightarrow P \rightarrow R.$$

- Example:
 - If the sun shine then we go to UNM then if we go to UNM then we are happy then if the sun shine then we are happy



- Here is the proof in Lean (C for compose)

```
theorem C : (P → Q) → (Q → R) → P → R :=  
begin  
  assume p2q,  
  assume q2r,  
  assume p,  
  apply q2r,  
  apply p2q,  
  exact p,  
end
```

[Try it!](#)



- First of all, this corresponds to:

$$(P \rightarrow Q) \rightarrow ((Q \rightarrow R) \rightarrow (P \rightarrow R))$$

- After the three assume we are in the following state:

```
P Q R : Prop,  
p2q  : P → Q,  
q2r  : Q → R,  
p    : P  
⊢ R
```



- Now use an implication.
- Clearly it is *q2r* which can be of any help because it promises to show *R* given *Q*.
- Hence once *apply q2r*, it is left to show *Q*:

```
P Q R : Prop,  
p2q  : P → Q,  
q2r  : Q → R,  
p    : P  
⊢ Q
```

- The next step is to use *apply p2q* to reduce the goal to *P* which can be shown using *exact P*.



- Note that there are two kinds of steps in these proofs:
- *assume h* means that prove an implication $P \rightarrow Q$ by assuming P (and this assumption is called h) and then proving Q with this assumption.
- *apply h* if an implication $h : P \rightarrow Q$ is assumed and the current goal matches the right hand side, then use this assumption to reduce the problem to showing P (whether this is indeed a good idea depends on whether it is actually easier to show P).



- Give yourself a big clap.
- You have survive until here. ◀◀
- Any question?



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Next lecture...

- Propositional Logic Part 2
 - More logic theory, more proving, more tactics...



“Imagination is more important than knowledge.
For knowledge is limited, whereas imagination
embraces the entire world, stimulating progress,
giving birth to evolution.”

– Albert Einstein

Hope you all enjoy this lecture!